



**A Course End Project Report on**  
**Electronic Voting Machine Using STM32**

**A8427- Microcontrollers Laboratory**

**Submitted by**

**Batch No.6**

**B Pranav Paartha    Roll No - 23881A0472**

**G Vinay                Roll No - 23881A0488**

**G Akshaya             Roll No - 23881A0489**

**Course Facilitator**

**Dr. R. Phani Vidyadhar**  
**Assistant Professor**

**Dr. J. Swetha Priyanka**  
**Assistant Professor**

**Department of Electronics and Communication Engineering**

**Vardhaman College of Engineering, Hyderabad**

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## Abstract

This project presents a simple, cost-effective, and secure implementation of an electronic voting machine (EVM) using an STM32 microcontroller programmed with STM32CubeIDE. The system facilitates accurate vote registration for multiple candidates through dedicated push buttons, ensuring a streamlined voting process. An embedded C program governs the voting mechanism, incorporating a software-based lock to prevent multiple votes within a single session, thereby enhancing voting integrity. Real-time results, including candidate-wise vote counts and the winner, are displayed on an LCD board. The project leverages a hardware setup comprising a breadboard, jumper wires, push buttons, and an LCD board, emphasizing practical STM32-based system design. It focuses on digital input processing, memory management, and output display control, offering hands-on experience in embedded systems development. The implementation serves as a foundation for future enhancements, such as vote encryption, biometric authentication, or integration with advanced interfaces.

**Keywords:** Electronic Voting Machine, STM32, Embedded C, STM32CubeIDE, LCD Board, Secure Voting

# 1 Introduction

Electronic voting machines (EVMs) have revolutionized electoral processes by providing faster, more accurate, and secure vote collection compared to traditional paper-based systems, which are prone to errors and tampering. This project develops a secure EVM using an STM32 microcontroller interfaced with a breadboard, jumper wires, push buttons, and an LCD board, programmed using STM32CubeIDE. Each candidate is assigned a dedicated push button for vote registration, and a software-based lock mechanism ensures that only one vote is recorded per session, safeguarding the voting process. The LCD board displays real-time vote counts and the winner, offering a user-friendly interface. This implementation provides practical experience in embedded C programming, STM32 microcontroller interfacing, and hardware integration, while demonstrating the feasibility of a low-cost, reliable EVM. The project also aims to foster understanding of secure system design and its application in democratic processes, with potential for scalability and enhancements.

## 2 Literature Review

Extensive research on EVMs underscores the importance of security, reliability, and usability in modern electoral systems. Advanced microcontrollers like the STM32, known for their high performance and versatility, are widely used in EVM designs due to their robust processing capabilities and support for peripherals. Studies, such as those by *Programming and Customizing the STM32 Microcontroller* by Elecia White, highlight the STM32's suitability for real-time applications, including input handling and display interfacing. Additional references, such as *Microprocessors and Interfacing* by D. V. Hall, emphasize the need for mechanisms to prevent multiple voting, ensure accurate vote tallying, and provide clear result displays. Hardware setups using breadboards, jumper wires, and LCD boards, as described in STM32-based project reports, enable rapid prototyping and testing. Academic projects often incorporate software-based lock mechanisms using flags or interrupts, as seen in *Embedded Systems Design* by A. S. Tanenbaum. This project builds on these principles, implementing a secure EVM with push buttons for input, a software lock for vote integrity, and an LCD board for output, validated through practical hardware testing and STM32CubeIDE programming.

## 3 Objectives

- Design and implement a secure EVM using an STM32 microcontroller, breadboard, jumper wires, push buttons, and an LCD board.
- Assign dedicated push buttons for candidate vote registration to ensure intuitive user interaction.
- Develop a software-based lock mechanism to prevent multiple votes per session, enhancing security.
- Accurately count and store votes for each candidate in the STM32's memory.
- Display real-time voting results and the winner on an LCD board for transparency.
- Gain hands-on experience in embedded C programming and STM32CubeIDE development.
- Understand STM32 input/output handling, hardware interfacing, and secure system design principles.

- Explore potential for future enhancements, such as encryption or advanced user authentication.

## 4 Methodology

### 4.1 Block Diagram

The system comprises the following components:

- **Push Buttons:** Dedicated keys for each candidate to register votes, ensuring ease of use.
- **STM32 Microcontroller:** Core processing unit that handles input detection, vote counting, and lock management.
- **Lock Mechanism:** Software-based flag to prevent multiple votes in a single session.
- **Vote Counter:** Internal memory registers to track votes per candidate.
- **LCD Board:** Displays real-time vote counts, lock status, and winner information.

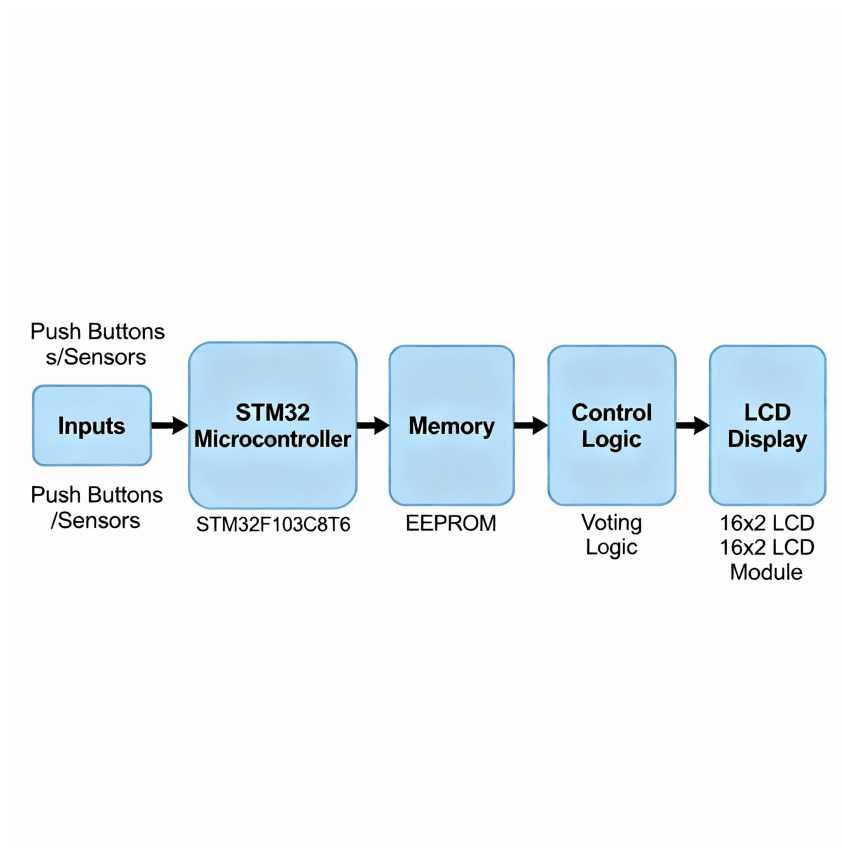


Figure 1: Block Diagram of Secure EVM

### 4.2 Circuit Diagram and Components

#### 4.2.1 Components Used

- **STM32 Microcontroller:** Processes inputs and controls the system.
- **Breadboard:** Provides a platform for circuit assembly and prototyping.

- **Jumper Wires:** Facilitate connections between components.
- **LCD Board:** Displays voting results and system status.
- **Push Buttons:** Serve as input devices for vote registration.

### 4.3 Working Principle

1. The voter presses a push button corresponding to their chosen candidate.
2. The STM32 microcontroller detects the button press via its GPIO pins and increments the respective candidate's vote count.
3. A software lock flag is set to prevent further votes in the same session, ensuring one vote per user.
4. Vote counts are updated in real-time and displayed on the LCD board, showing candidate-wise tallies.
5. At the end of the polling session, the system compares vote counts to determine and display the winner on the LCD board.

### 4.4 Software Implementation

The software is developed using STM32CubeIDE and embedded C, with the following key functionalities:

- **Input Handling:** Configures STM32 GPIO pins to monitor push button inputs, with debouncing to ensure accurate detection.
- **Vote Counting:** Maintains vote counters in the STM32's memory, incrementing them for valid button presses.
- **Lock Control:** Implements a software flag that locks the system after a vote, reset manually via a designated button or automatically after a session timeout.
- **Output Display:** Interfaces with the LCD board (e.g., 16x2 LCD) using a library like HD44780 to display vote counts, lock status, and the winner.
- **Hardware Testing:** The program is written, debugged, and uploaded to the STM32 via STM32CubeIDE, with testing performed on a breadboard setup to verify functionality of push buttons and LCD board.

## 5 Results and Discussion

- **Setup:** The EVM is implemented on an STM32 microcontroller connected to a breadboard, with jumper wires linking push buttons and an LCD board. The system is programmed and tested using STM32CubeIDE.
- **Output Display:** The LCD board displays real-time information, including vote registration, candidate-wise vote counts, lock status, and the winner at the session's end.
- **Voting Behavior:**
  - **Random Mode:** Push buttons are used to simulate random voting patterns, testing the system's ability to count votes and enforce the lock mechanism (Figure 2).

- **Manual Mode:** Votes are registered through individual push button presses, validating accurate input detection and vote tallying (Figure ??).
- **Lock Mechanism:** The software lock effectively prevents multiple votes until the system is reset, ensuring voting integrity.
- **Winner Determination:** The system accurately identifies the candidate with the highest vote count and displays the result on the LCD board.
- **Discussion:** The hardware setup, combined with robust software, confirms reliable input handling, vote counting, and LCD output. The system demonstrates a practical and secure EVM architecture, with the STM32 efficiently managing inputs and outputs. Challenges, such as button debouncing and LCD interfacing, were addressed through careful programming and testing. The project highlights the potential for real-world EVM applications and provides a scalable framework for enhancements like secure data storage or network connectivity.

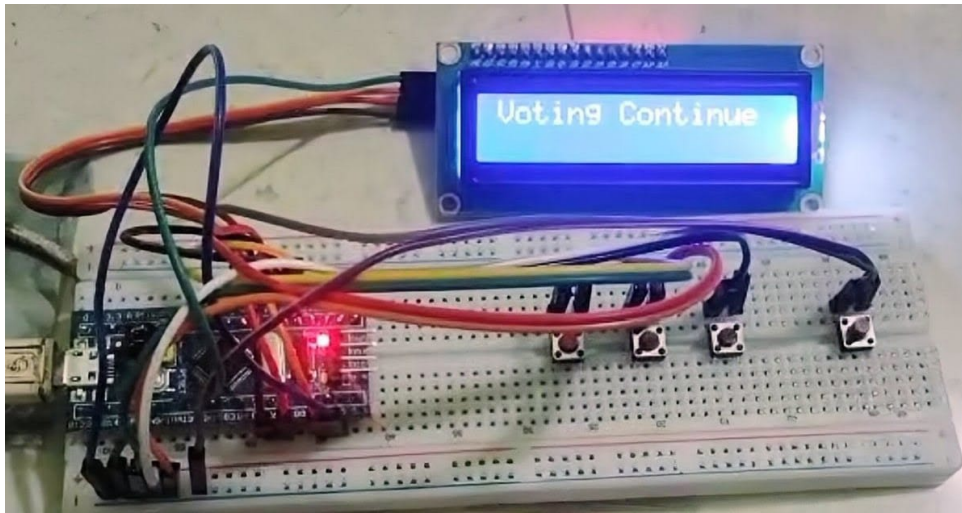


Figure 2: LCD Board Output for Random Voting

## 6 Mapping POs and SDGs

### 6.1 Program Outcomes (POs) Mapping

Table 1: Mapping of Program Outcomes to Project

| PO No. | Program Outcome       | Relevance                                                                               |
|--------|-----------------------|-----------------------------------------------------------------------------------------|
| PO1    | Engineering Knowledge | Applied STM32 programming, GPIO configuration, and hardware interfacing for EVM design. |
| PO2    | Problem Analysis      | Addressed vote integrity through software lock mechanisms and button debouncing.        |
| PO3    | Design/Development    | Developed embedded C code for vote counting, lock control, and LCD board output.        |
| PO4    | Investigations        | Conducted extensive testing of voting scenarios to validate system functionality.       |
| PO6    | Engineer and Society  | Designed a secure EVM to support transparent and fair democratic processes.             |

### 6.2 Sustainable Development Goals (SDGs) Mapping

Table 2: Mapping of SDGs to Project

| SDG No. | Goal                 | Relevance                                                                      |
|---------|----------------------|--------------------------------------------------------------------------------|
| SDG 4   | Quality Education    | Enhances learning through hands-on STM32 programming and hardware integration. |
| SDG 9   | Industry, Innovation | Develops innovative, low-cost voting technology using accessible components.   |
| SDG 16  | Peace, Justice       | Promotes transparent and secure voting systems for fair elections.             |



## 7 Conclusion

This project successfully developed a secure and efficient electronic voting machine using an STM32 microcontroller, breadboard, jumper wires, push buttons, and an LCD board. The system supports both manual and random voting modes, with accurate vote counting and a robust software-based lock mechanism to prevent multiple votes. Results are displayed in real-time on the LCD board, providing clear visibility of vote counts, lock status, and the winner. The implementation demonstrates a practical approach to embedded systems design, leveraging STM32CubeIDE for programming and a breadboard-based hardware setup for testing. The project not only validates the feasibility of a low-cost, secure EVM but also provides a scalable platform for future enhancements, such as integrating encryption, biometric authentication, or wireless connectivity. This work contributes to the advancement of secure voting technology and offers valuable insights into STM32-based system development.

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